



Pien-Tze-Huang ameliorates hepatic fibrosis via suppressing NF- κ B pathway and promoting HSC apoptosis

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ABSTRACT

Ethnopharmacological relevance: Pien Tze Huang (PZH), a Chinese herbal formula, has various forms of pharmacological activity including anti-inflammation, liver protection and anti-tumor.

Aim of the study: To investigate the anti-hepatic fibrosis effect of PZH and its potential mechanisms on experimental animal model of hepatic fibrosis and hepatic stellate cell (HSC).

Material and methods: Rats were intraperitoneally administered with CCl₄ to induce hepatic fibrosis and were simultaneously treated with PZH. Liver pathology were observed by hematoxylin and eosin (H&E) staining and Masson staining. Serum pro-inflammatory cytokines were determined by enzyme-linked immunosorbent assay (ELISA) kits. Moreover, the effects of PZH on HSC proliferation and apoptosis were assessed via MTT, flow cytometry and immunofluorescence analysis. Nuclear factor- κ B (NF- κ B) pathway activation and Bcl-2 family members were evaluated by reverse transcription quantitative polymerase chain reaction (real-time PCR) and western blotting.

Result: PZH significantly alleviated CCl₄-induced liver injury, inflammation and fibrogenesis in rats. PZH also markedly decreased the production of hepatic-fibrosis biomarker, including ALT, AST, IV collagen and PCIII (Procollagen III), as well as inflammatory cytokines such as IL-1 β , IL-6 and TNF- α . Importantly, PZH strongly inhibited HSC proliferation correlated with cell apoptosis induction as evidenced by modulating Bcl-2 family members and caspase activity. Moreover, PZH administration significantly increased the expression I κ B- α , an inhibitor of NF- κ B and suppressed expression of anti-apoptotic genes (Bcl-2, etc.). Collectively, these results suggested that PZH could promote HSC apoptosis through inhibiting NF- κ B signaling pathway.

Conclusion: Our study demonstrates that PZH ameliorates hepatic fibrosis and inflammation, chiefly through suppressing the NF- κ B pathway and promoting HSC apoptosis. PZH is more likely to be a promising antifibrotic agent in chronic disease.

1. Introduction

Hepatic fibrosis is a pathophysiological consequence of wound-healing response to chronic injury caused by various etiologies, mainly involving alcohol abuse, hepatitis viral infection, nonalcoholic steatohepatitis, autoimmune and cholestatic liver diseases. Chronic liver injury and the associated wound-healing response contributing to imbalance between the synthesis and degradation of extracellular matrix (ECM) in liver tissue are tightly related to the development of fibrosis,

cirrhosis and hepatocellular carcinoma (HCC) (Asrani et al., 2018; Parola and Pinzani, 2018). Notably, more than 80 percent of HCCs, the final stage of this process, occur from patients with hepatic fibrosis and cirrhosis (Luedde and Schwabe, 2011). Over the past 3 decades, there have been accumulating documents indicating that hepatic fibrosis is reversible and multiple mechanisms underlying hepatic fibrogenesis have been uncovered (Rockey, 2009).

A great range of studies have demonstrated that hepatic stellate cell (HSC) is a critical effector cell in liver fibrogenic process (Guo et al.,

Abbreviations: PZH, Pien Tze Huang; H&E, Hematoxylin and Eosin, Masson; ELISA, Enzyme-linked immunosorbent assay; HSC, Hepatic stellate cell; NF- κ B, Nuclear factor- κ B; ECM, Extracellular matrix; HCC, Hepatocellular carcinoma; TNF- α , Tumor necrosis factor; IL-1, Interleukin; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide; I κ B- α , Inhibitor of κ B- α ; I κ B- β , I κ B kinase β

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2015; Huang et al., 2017). Upon liver injury, this cell undergoes a transformation from a “quiescent” to an “activated” state. The principal characteristic of the activated HSC is the synthesis and secretion of a large amount of ECM, leading to over-deposition of fibrous tissue. Additionally, the activated cell has a stronger proliferative ability and also releases proinflammatory and profibrogenic cytokines (Zhang et al., 2016). More importantly, there is significant increase in HSCs apoptosis during recovery from hepatic fibrosis (Huang et al., 2017). Thus, it can be seen impeding the activation of HSC or inducing its apoptosis will probably be a therapeutic approach against hepatic fibrogenesis and elucidating these mechanisms are of great important in highlighting novel potential therapies (Angeles et al., 2009; Elsharkawy et al., 2005).

Nuclear factor- κ B (NF- κ B) is a transcriptional regulator of genes involved in immunity, inflammatory response and cell survival. Emerging evidences reveal that it acts as a master regulator of inflammation and cell death in the development of hepatocellular injury, hepatic fibrosis and HCC (Mann and Smart, 2002). When liver injury is continuous or chronic, NF- κ B is activated and regulates multiple essential functions in a variety of parenchymal and nonparenchymal liver cells. At the initial stage of hepatic injury, apoptosis and necrosis of hepatocytes activates Kupffer cells with subsequent activation of NF- κ B, resulting in releasing massive proinflammatory cytokines, the most important being tumor necrosis factor (TNF)- α and interleukin (IL)-1 (Bahçecioğlu et al., 2004). These in turn act on their cognate receptors on both hepatocytes and HSCs. In hepatocytes, appropriate NF- κ B activation results in enhanced cell survival and anti-apoptosis, whereas too much or too little NF- κ B activation may worsen inflammation and further damage to the cell or lack effective protection from cell death. In HSCs, injury-induced activation of the cell is again through activation of NF- κ B, which is a pivotal regulator of HSC survival and fibrogenic responses in chronic liver disease (Elsharkawy and Mann, 2007; Luedde and Schwabe, 2011).

Pien Tze Huang (PZH), a famous traditional Chinese formula with a history of more than 450 years since the Ming Dynasty, has definite clinical efficacy in anti-inflammation and liver protection, and used in treatment for fatty liver, chronic hepatitis, inflammatory disease and even cancer (Qi et al., 2017; Yang et al., 2018). The ingredients of PZH are mainly composed of *Panax notoginseng* (Burk.) F. H. Chen, *Calculus Bovis*, and *Moschus*. The previous study of PZH showed that it can ameliorate hepatic injury and inflammation in rats with carbon tetrachloride (CCl₄)-induced hepatitis model, and prevent hepatocellular apoptosis (Zhao et al., 2017). Recently, our preliminary experiments have shown that PZH significantly inhibited the proliferation of HSC. These indicate that PZH may be a potential therapeutic agent to hepatic fibrosis. Therefore, in the present study, we employed an animal model with hepatic fibrosis induced by CCl₄ to investigate the anti-hepatic fibrosis effect of PZH and hypothesized that its mechanism may be through the regulation of NF- κ B pathway and HSC survival, eventually providing scientific basis for the clinical application of this drug.

2. Materials and methods

2.1. Materials

RPMI Medium 1640 basic (1640), fetal bovine serum (FBS), Trypsin and penicillin-streptomycin were all purchased from Hyclone (Logan, UT, USA). Dimethyl sulfoxide (DMSO) and 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) were purchased from Sigma-Aldrich (St. Louis, MO, USA). The Annexin V/propidium iodide (PI) apoptosis assay kit was obtained from KeyGEN BioTECH (Nanjing, China). Antibodies to NF- κ B (#8246), Bcl-2 (#2876), Bax (#5023S), IKK- α (#4812) and GAPDH (#2118) were purchased from Cell Signaling Technology, Inc. (Danvers, MA, USA). Antibodies to IKK- β (#Ab124957) was purchased from Abcam (Cambridge, Cambs, UK). Antibodies to p-Bad (#AB55025) was purchased from Sangon Biotech

Co., Ltd (Shanghai, China). The secondary antibodies were from Xiamen Lulong Biotech Development Co., Ltd. (Xiamen, China). Polyvinylidene fluoride (PVDF) membranes were from Merck KGaA (Darmstadt, Germany). All other reagents were from the Beyotime Institute of Biotechnology (Nanjing, China) unless otherwise stated.

2.2. Preparations of PZH

PZH was obtained from and authenticated by the sole manufacturer Zhangzhou Pien Tze Huang Pharmaceutical Company Limited, China (Chinese FDA approval number: Z35020242). Stock solutions of PZH were prepared just before use by dissolving the PZH powder in dimethyl sulfoxide (DMSO) to a concentration of 50 mg/mL. The working concentrations of PZH were made by diluting the stock solution in the culture medium.

2.3. Animals and treatment

Male Sprague-Dawley rats with a body weight of 250 ± 20 g were purchased from Shanghai SLAC Laboratory Animal Co., Ltd. (SCXK2017-0005, Shanghai, China). Briefly, forty were randomly divided into four groups (ten rats per group) as follows: normal control group, CCl₄ model group and PZH-treatment group and colchicine-treatment group as a positive control. Except the normal control group, rats from the last three groups were intraperitoneally administered with 1 mL/kg CCl₄ (mixed 1:1 in peanut oil) twice a week for 8 weeks. Simultaneously, rats from PZH-treatment group was administered with PZH at an oral dose of 0.15 g/kg once a day, and colchicine-treatment group was administered with colchicine at an oral dose of 1 mg/kg once a day, while the CCl₄ model group received equivalent amounts of normal saline. At an endpoint, all animals were sacrificed and blood and liver samples were obtained for further analysis. All animal experiments were carried out by guidelines of the National Institute of Health for the Care and Use of Laboratory Animals (NIH Publication 80-23, revised in 1996) and with approval of the Fujian Institute of Traditional Chinese Medicine Animal Ethics Committee (Fuzhou, China), and the protocols were carried out according to the Guidelines for Animal Experimentation of Fujian University of Traditional Chinese Medicine (Fuzhou, China).

2.4. Enzyme-linked immunosorbent assay

The expression levels of ALT, AST, IV collagen, PCIII, IL-1 β , IL-6, TNF- α were detected using an Enzyme-linked immunosorbent assay (ELISA) kit. Procedures were performed according to the manufacturer's protocol. Ultimately, the absorbance at 450 nm was measured using a microplate reader.

2.5. Hematoxylin and eosin staining and masson staining

Fresh liver samples were embedded in paraffin for histological examination using hematoxylin and eosin (H&E) staining and Masson staining. Sections with 5 μ m thick were cut and prepared for further experiments. Morphological changes were observed under a light microscope with a digital camera. At the same time, we graded the degree of hepatic fibrosis by Masson staining. Grading criteria for hepatic fibrosis: no fibrosis, no necrosis of hepatocytes, 0 point; fibrous tissue hyperplasia, fibrosis limited in portal area of porta hepatis, 1 point; hepatic fibrosis occurs around the portal area of porta hepatis, the fiber spacing formed, and liver lobule structure retained, 2 points; fibrous septa with liver lobular structure disorder, but no obvious cirrhosis, 3 points; false hepatic lobule formation, early cirrhosis, 4 points.

2.6. Cell culture and treatment

HSC-T6 cells (Sangon Biotech Co., Ltd, Shanghai, China) were

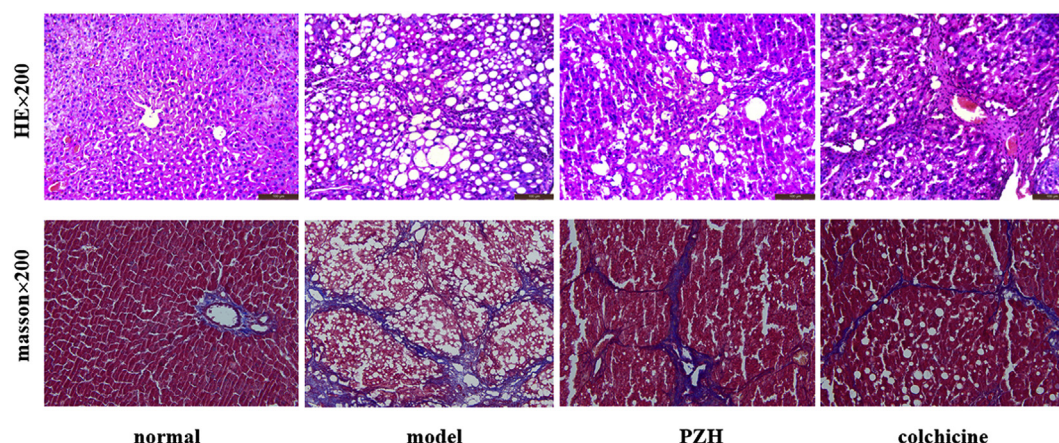


Fig. 1. Effect of PZH on hepatic histological changes. Representative images of Hematoxylin-eosin (H&E) staining and masson staining from normal control group, CCl₄ model group, PZH-treatment group and colchicine-treatment group were shown at $\times 200$ magnification, respectively.

Table 1

Hepatic fibrosis rat grade score ($\bar{X} \pm S$, $n = 6$).

Group	Degree of liver fibrosis
Normal group	0 ± 0
Model group	$2.67 \pm 0.82^{##}$
PZH group	$1.67 \pm 0.52^*$
Colchicine group	$1.33 \pm 0.52^*$

$p < 0.05$, ## $p < 0.01$ vs. the normal group and * $p < 0.05$,

** $p < 0.01$ vs. the CCl₄ model group.

cultured in 1640 medium containing 10% FBS and 100U/mL penicillin-streptomycin at 37°C in a humidified incubator with 5% CO₂. After seeding onto 96-, 24- or 6-well plates for 24 h, the cells were treated with or without various concentrations of PZH (150, 200, 250, 300 μ g/mL) for 48 h for subsequent experiments. The control cells were not treated with PZH as the vehicle control.

2.7. Determination of cell viability

Cell viability of HSC-T6 was detected by MTT assay. Briefly, Cells were seeded into 96-well plates at a density of 5×10^3 cells/well and then administrated with different concentrations of PZH. Following that, 10 μ L MTT (5 mg/mL) were added to each well for an additional 4 h of incubation at 37 °C. The MTT solution was then replaced with DMSO (100 μ L/well) to dissolve the formazan precipitate. The absorbance at 570 nm was measured using a microplate reader (Infinite M200 Pro; Tecan, Männedorf, Switzerland). The assay was performed in triplicate and each group was eight wells. Cell viability was calculated via the ratio of absorbance values between the administrated group and the control group.

2.8. Immunofluorescence assay

After administrated with PZH as described above, the culture-medium was removed and thence cells were fixed with 4% paraformaldehyde solution for 15 min at room temperature, followed by washed three times with PBS. Before rinsed with PBS three times, the cells were stained with DAPI solution for 10 min. Finally, the apoptotic cells were observed under a fluorescence microscope (Leica Microsystems, Germany) at $\times 200$ magnification. The experiment was performed in triplicate.

2.9. Flow cytometry analysis

Following treatment with consecutive concentrations of PZH, the

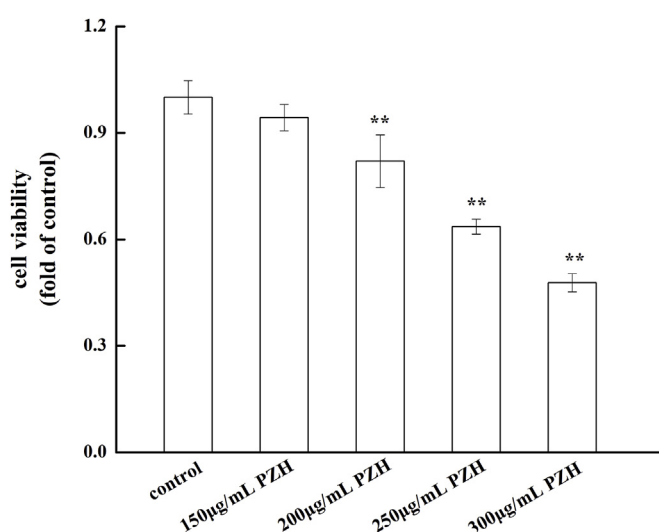


Fig. 2. Effect of PZH on cell viability of HSC-T6. The results of MTT assay were expressed as percentages of cells treated with consecutive concentrations of PZH (150, 200, 250, 300 μ g/mL) for 24 h compared with the control group (* $p < 0.05$, ** $p < 0.01$).

cells were harvested and the apoptotic cells were determined by flow cytometric analysis utilizing Annexin V/PI kit. The procedure of cell staining was performed according to the manufacturer's instructions. Finally, the samples were then detected by a flow cytometer with an excitation wavelength of 488 nm and an emission wavelength of 530 nm and the apoptosis rate was analyzed with BD Cell Quest™ software (Becton-Dickinson, Bedford, MA, USA). The experiment was repeated three times.

2.10. Real-time PCR analysis

Total RNA was extracted from cultured cells or tissue samples using TRizol reagent (Invitrogen, Barcelona, Spain) and then was reverse transcribed using the PrimeScript® RT reagent kit (Takara Bio, Inc., Otsu, Japan) for first strand complementary DNA (cDNA) synthesis. Real-Time quantitative PCR was then carried out using a Power SYBR®-Green PCR Master mix (Thermo Fisher Scientific, Inc., Shanghai, China) with the 7900 Real-Time PCR system (Applied Biosystems, Inc., Foster City, CA, USA). The expression level was normalized against endogenous GAPDH. Primer sequences used in the reactions were as follows: GAPDH reverse 5'- GCTTACCACCTTCTT GATGTC-3' and forward primer 5'- TGAACGGGAAGCTCACTGG-3'. Bax reverse primer, 5'-

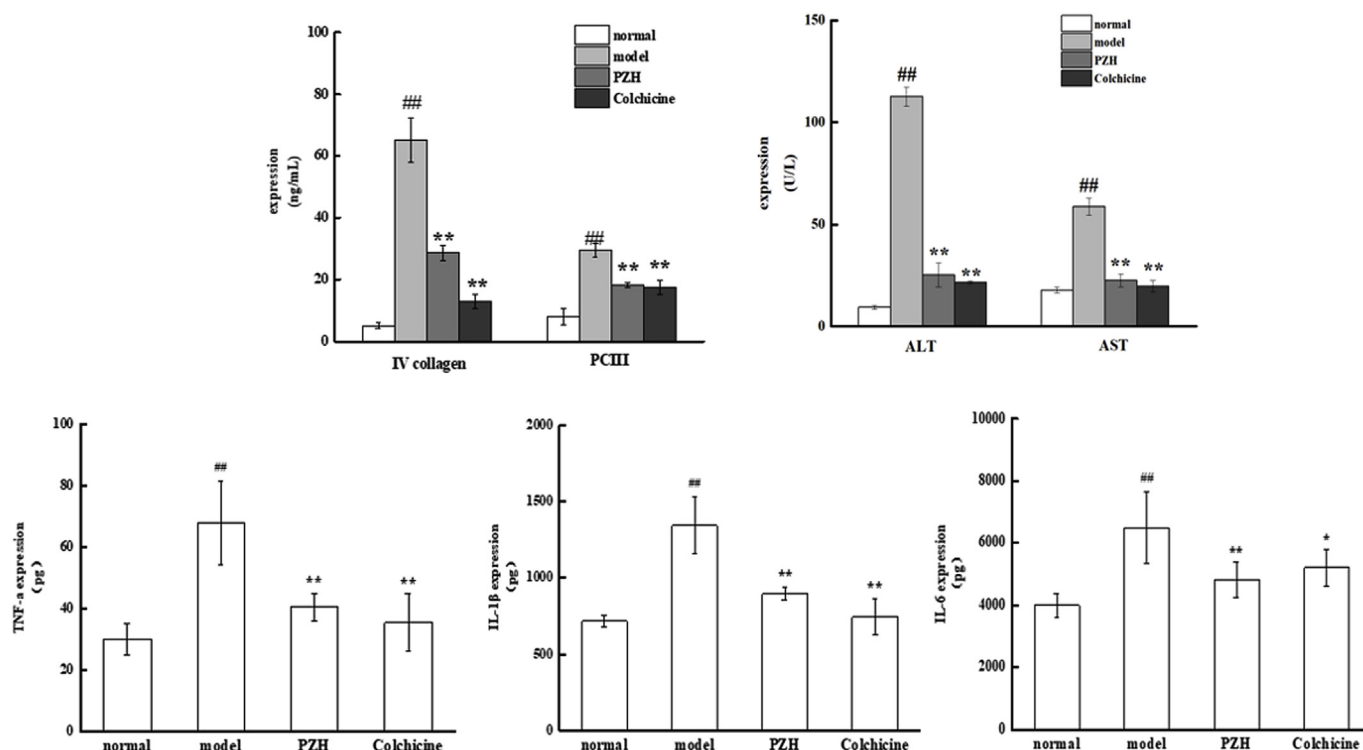


Fig. 3. Effect of PZH on pro-inflammatory cytokines including ALT, AST, IV collagen, PCIII, IL-1 β , IL-6 and TNF- α in serum. ELISA assay was employed and data was expressed as mean $\pm$ SD of three independent experiments. #p < 0.05, ##p < 0.01 vs. the normal control group and *p < 0.05, **p < 0.01 vs. the CCl₄ model group.

GAT CAG CTC GGG CAC TTT AG-3' and forward primer, 5'-TGC AGA GGA TGA TTG CTG AC-3'; Bcl-2 reverse primer, 5'-ATG CCG GGT CAG GTA CTC AG-3' and forward primer, 5'-GGT GGT GGA GGA ACT CTT CA-3'.

2.11. Western blot analysis

Cultured cells or tissue samples were extracted with RIPA lysis buffer containing protease inhibitor cocktail for 30 min on ice, and were subsequently centrifuged at 12,000 rpm for 15 min at 4 °C. The supernatant was collected and the protein concentration was determined using the BCA method. The protein was mixed with loading buffer and heated at 100 °C for 10 min, followed by separated in on 10% and 12% SDS acrylamide gels via electrophoresis and transferred to polyvinylidene fluoride membranes by an electric transfer system. The membranes were blocked with 5% skim milk in TBS buffer containing 0.1% Tween-20 for 1 h at room temperature and then incubated with antibodies against α -SMA (1:1000), NF- κ B (1:1000), Bcl-2 (1:1000), Bax (1:1000), p-bad (1:5000), I κ B- α (1:1000), I κ B- β (1:2000), and GAPDH (1:1,000) overnight at 4 °C. Subsequently, the membranes were incubated with appropriate secondary antibody for 2 h at room temperature. Finally, the blots were evaluated using the ECL western detection reagents on Image Lab analysis software (Bio-Rad Laboratories, Inc., Hercules, CA, USA). GAPDH was used as an internal control. Three repeats of the experiments were performed.

2.12. Statistical analysis

Statistical analysis was carried out using SPSS 21.0 for Windows. One-way analysis of variance (ANOVA) was used to analyze the results expressed as mean $\pm$ SD. Differences were considered significant when the P value was less than 0.05.

3. Results

3.1. PZH alleviated liver lesion and fibrogenesis in rats

In this study, the administration of PZH could significantly alleviate hepatic injury and fibrogenesis induced by tetrachloride in rats, which was consistent with our previous research results (Another paper was published). As shown in Fig. 1, liver tissues in the model group showed obvious vacuolar degeneration and necrosis of hepatocytes, infiltration of a large number of inflammatory cells as well as deposition of collagen, destruction of normal hepatic lobule structure and even reconstruction of hepatic lobule structure. In both treatment groups, vacuolar degeneration of liver tissues was significantly reduced, inflammatory cells and fiber hyperplasia were also significantly reduced. As shown in Table 1, the two drug groups, Pien-Tze-Huang and colchicine, significantly reduced the degree of hepatic fibrosis in rats compared with the model group. These results showing PZH had a great beneficial effect.

3.2. PZH inhibited HSC-T6 cell viability

To investigate cell viability of HSC-T6 cell treated with consecutive concentrations of PZH, an MTT method was employed. As shown in Fig. 2, treatment with PZH (150, 200, 250 and 300 μ g/mL, respectively) for 24 h significantly reduced cell viability in a dose-dependent manner. The 50% inhibitory concentration (IC) was 280 ± 6.8 μ g/mL.

3.3. PZH suppressed hepatic-fibrosis biomarker, inflammatory cytokines and NF- κ B pathway

To explore the potential anti-fibrosis role of PZH in liver, we evaluated the expression of hepatic-fibrosis biomarker, including ALT, AST, IV collagen, PCIII. The results showed that the biochemical markers of hepatic fibrosis in the model group were significantly higher than those

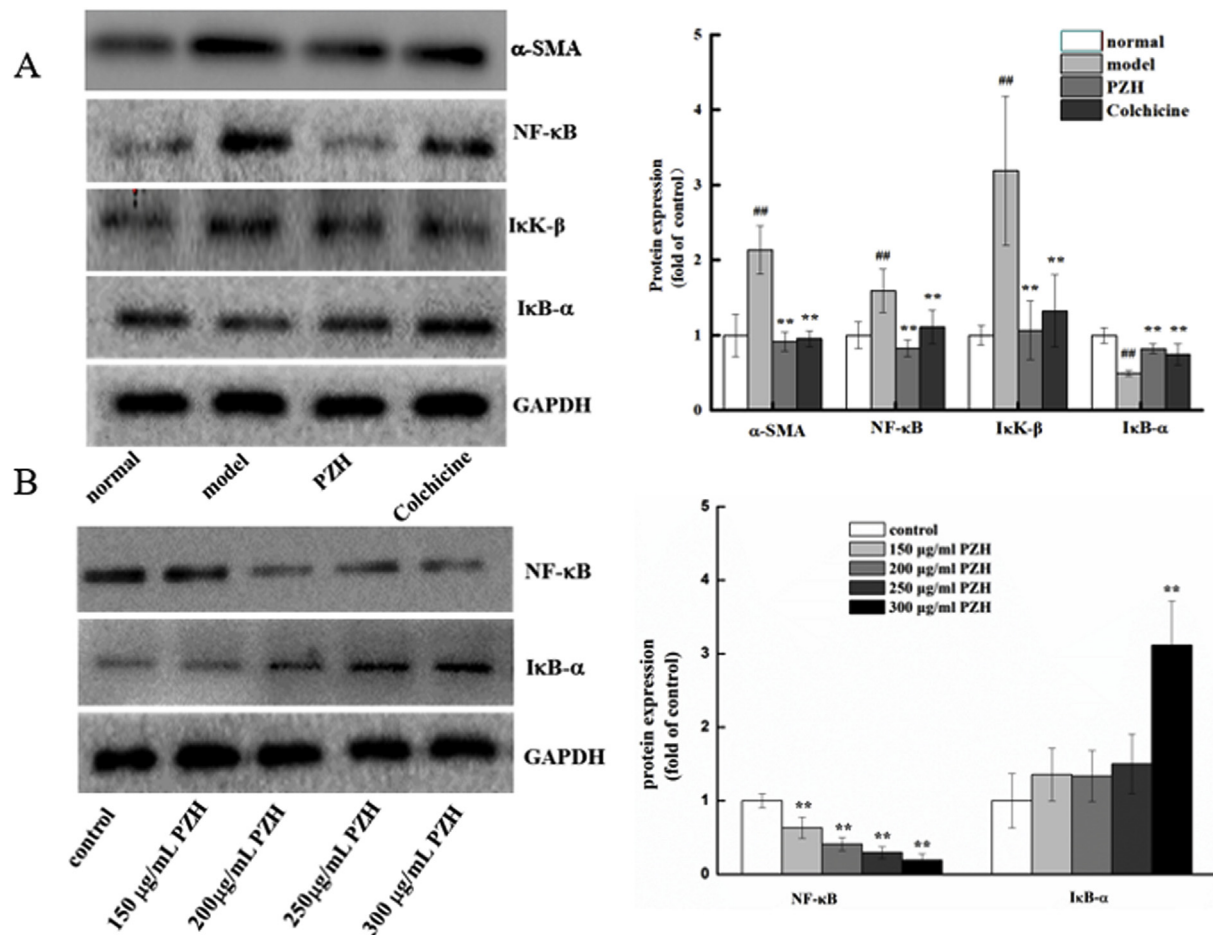


Fig. 4. Effect of PZH on NF-κB pathway in vivo and vitro. (A) The expression of NF-κB, IκB and IκK protein in liver tissues was detected by western blot. Expression was normalized against an endogenous control GAPDH and results expressed as mean $\pm$ SD of three independent experiments. ^{##} $p < 0.05$, ^{###} $p < 0.01$ vs. the normal control group and ^{*} $p < 0.05$, ^{**} $p < 0.01$ vs. the CCl₄ model group. (B) The expression of NF-κB, IκB and IκK protein in HSC was examined by western blot. ^{*} $p < 0.05$, ^{**} $p < 0.05$ vs. the control group.

in the normal group, and the abnormal expression of these indicators was significantly reduced after PZH drug treatment. pro-inflammatory cytokines including IL-1 β , IL-6 and TNF- α by ELISA. The results revealed that the levels of serum pro-inflammatory cytokines expression in the model group were significantly higher than that in the normal control group, while the abnormal levels of these inflammatory cytokines were reversed by PZH treatment (Fig. 3).

Firstly, α -SMA is a biomarker of activated HSC and reflects the degree of activation of HSC. The experiment had shown that PZH could significantly reduce the protein expression of α -SMA in rats with hepatic fibrosis. Since NF-κB is a key transcriptional factor which can regulate the expression of various pro-inflammatory mediators, NF-κB and its regulatory molecules such as inhibitor of κ B- α (I κ B- α) and I κ B kinase β (I κ K- β) were assessed in both liver tissues and HSC by Western blot. As shown in Fig. 4A, PZH administration significantly downregulated the expression of NF-κB and its kinase, I κ K- β , as well as upregulated the expression of I κ B- α contrasted to the model group. On the other hand, protein expression levels of NF-κB were remarkably down regulated with a concentration-dependent manner in PZH-treated hepatic stellate cells, conversely the expression of I κ B- α was elevated (Fig. 4B). These data suggested that PZH suppressed the production of inflammatory cytokines possibly via modulating the activation of NF-κB pathway in HSC.

3.4. PZH induced HSC apoptosis

The effect of PZH on HSC-T6 cell apoptosis was examined using flow

cytometry analysis. Annexin V is a Ca²⁺ dependent phospholipid-binding protein that has a special affinity for phosphatidylserine. In apoptotic cell, phosphatidylserine is translocated from the inner to the outer leaflet of the cytomembrane and thereby Annexin V can bind it. As shown in Fig. 5, LR region (PI-/Annexin V+) represented early apoptotic cells, and UR region (PI+/Annexin V+) represented late apoptotic cells. Compared with the control group, there was a significant increase in the ratio of apoptosis (including the early late apoptosis ratio) in a concentration-dependent manner in HSC-T6 cells after the treatment with PZH. We quantitatively analyzed the apoptotic rate of different groups simultaneously. The apoptosis ratios were: $6.33 \pm 0.75\%$, $8.86 \pm 0.57\%$, $11.3 \pm 0.81\%$, $33.28 \pm 8.02\%$, respectively. Consistent with the flow cytometry results, DAPI staining showed that the cell morphology in control group was normal, with a round or oval nuclei and uniform nuclear staining. Nevertheless, with the increase of the dose of PZH, the amounts of apoptotic cells increased, especially the 300 μ g/mL group, which reached the highest apoptotic number. When apoptosis occurred, the cells became thinner, nuclear shrinkage and staining deepened, and nuclear fragments increased, and apoptotic bodies appeared (Fig. 6). Taken together, these data demonstrated that PZH could induce HSC apoptosis.

3.5. PZH modulated expression of Bcl-2 family and Caspase-3 activity

To further investigate the effect of PZH on Bcl-2 family, a group of crucial apoptotic regulatory proteins, we chose an anti-apoptotic protein Bcl-2 and pro-apoptotic protein (Bax, p-bad) for assessment the

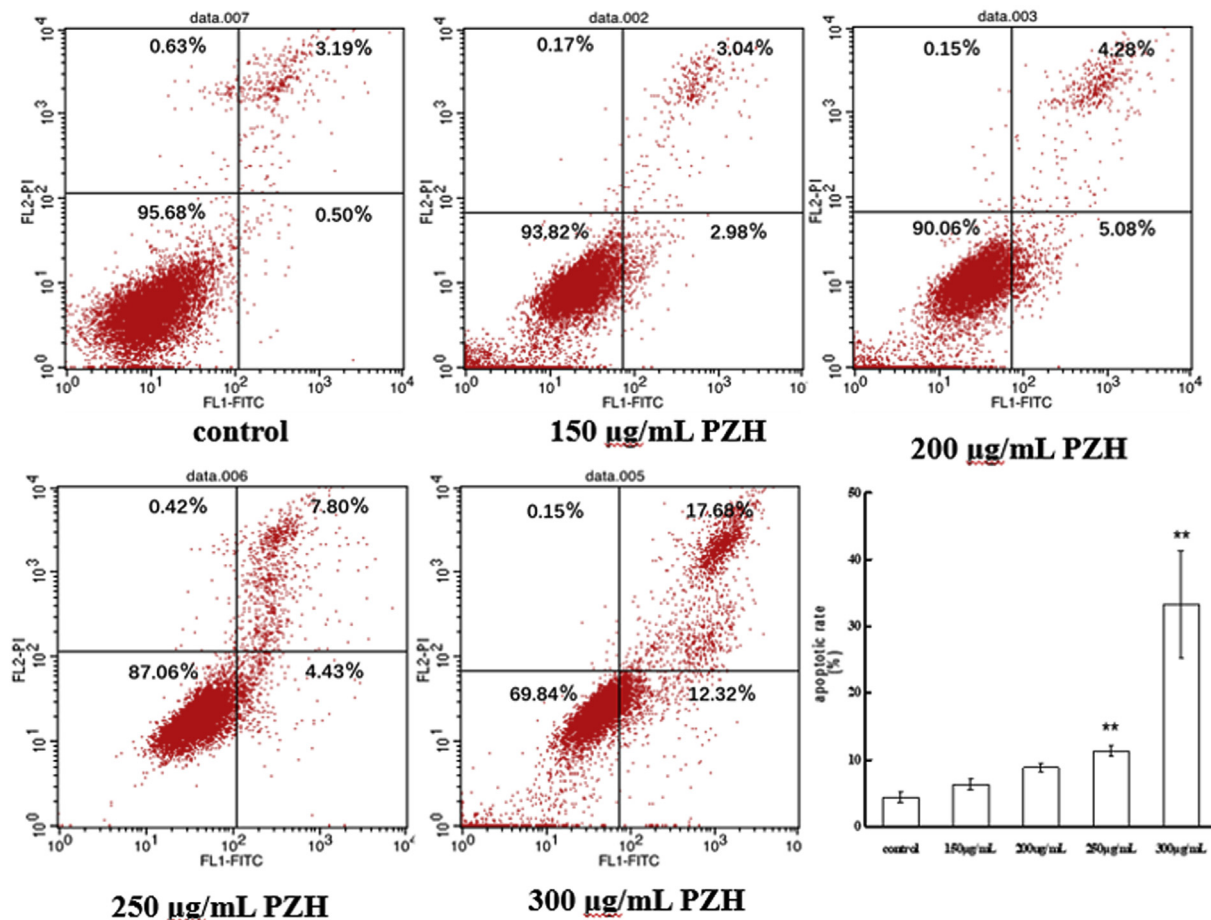


Fig. 5. HSC-T6 cell apoptosis was examined using flow cytometry analysis. Representative images from HSC treated consecutive concentrations of PZH were shown. Apoptosis rate was employed and data was expressed as mean $\pm$ SD of three independent experiments. * $p < 0.05$, ** $p < 0.01$ vs. the control group.

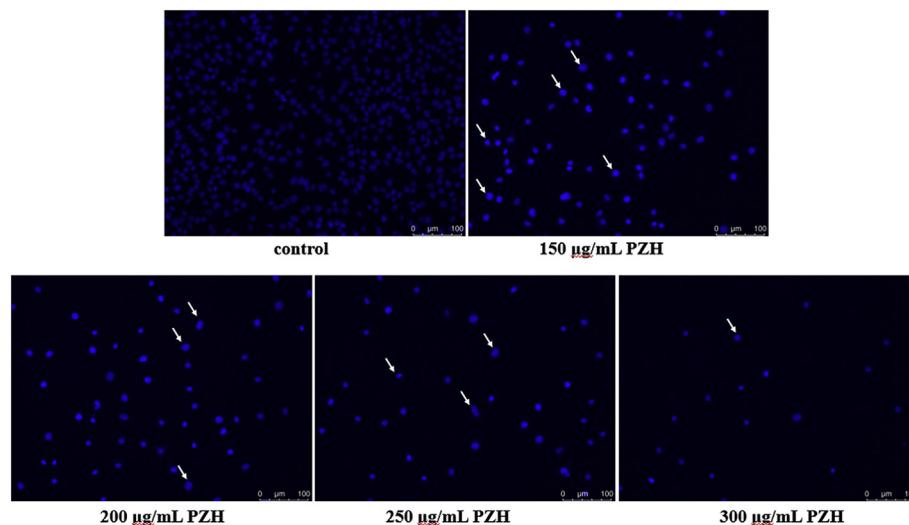


Fig. 6. DAPI staining showed the morphology of apoptotic cells. Cells treated with consecutive concentrations of PZH for 24 h and then stained with DAPI solution. Representative images were shown at $\times 200$ magnification. Apoptotic cells were marked with arrows.

expression levels. As shown in Fig. 7, real-time PCR and western bolt results revealed the expression of anti-apoptotic protein (Bcl-2) was downregulated in liver tissues from PZH treatment groups versus the control group, while the expression of pro-apoptotic protein (Bax, p-bad) was upregulated. In vitro, PZH could reduce the expression of Bcl-2 mRNA and protein, on the contrary, increase the expression of Bax

mRNA and protein in HSC (Fig. 8).

Moreover, Caspase-3, a key executioner of apoptosis, was also examined through fluorescence staining. Fig. 9 showed the PZH treatment group significantly increased the number of Caspase-3 positive cells contrast to the control group in a concentration-dependent manner. Collectively, these results implicated that PZH induces HSC apoptosis

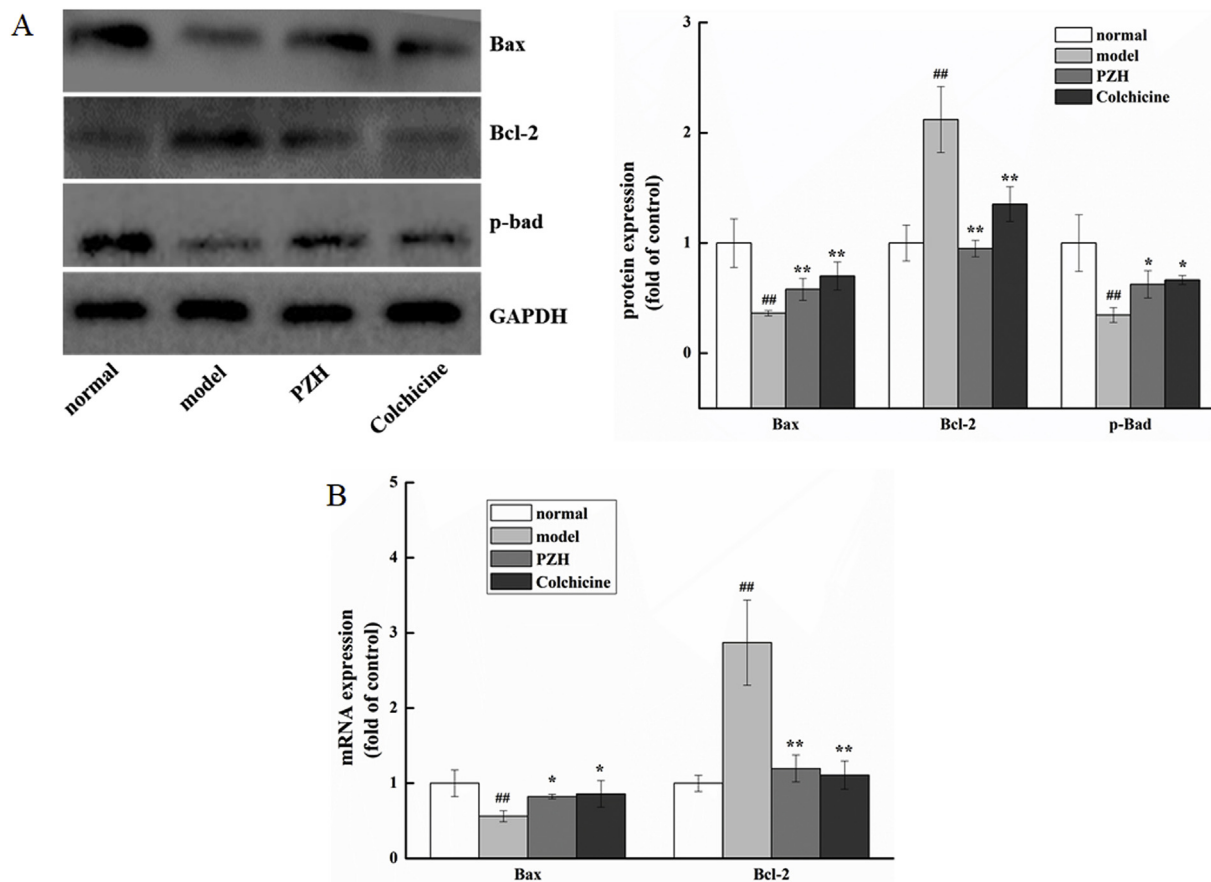


Fig. 7. Effect of PZH on Bcl-2 family in liver tissues of rats. (A) The expression of Bcl-2, Bax and p-bad protein was detected utilizing western bolt assay. (B) The expression of Bcl-2 and Bax mRNA was assessed by quantitative PCR assay. Expression was normalized against an endogenous control GAPDH and results were expressed as mean $\pm$ SD of three independent experiments. #p < 0.05, ##p < 0.01 vs. the normal control group and *p < 0.05, **p < 0.01 vs. the CCl₄ model group.

partially via the regulation the expression of Bcl-2 family proteins and caspase-3 activity.

4. Discussion

Hepatic fibrosis is a wound-healing response to chronic hepatic injury, leading to hepatic dysfunction and may even progress to liver cirrhosis and primary HCC. However, there are not yet any fibrosis-specific liver disease therapies that have been shown to be effective or approved for clinical use in chronic human liver disease (Friedman, 2015; Rudnick, 2017). Thus, further research will be necessary. As a traditional Chinese herbal preparation, PTH has been used as a folk remedy with heat-clearing, anti-inflammatory and detoxification effects for a long time. In modern medicine, PTH is principally applied to treat liver diseases, inflammation and cancer. Recently, it has been turned out that PTH can exert a hepatoprotective effect on various liver injury induced by CCl₄, alcohol or high-fat diet. Furthermore, the mechanism for its action is mainly associated with suppression of oxidative stress, inflammation and mitochondrial apoptosis, etc (Yang et al., 2018; Zhao et al., 2017). Additionally, Fei Qi et al. reported that PZH can effectively inhibit the growth of hepatocellular carcinoma cells, BEL-702, and induce cell apoptosis (Qi et al., 2017). Taken together, these findings suggest that PZH may be a promising protective agent in chronic liver diseases. Still, whether it has the potential to fight hepatic fibrosis is elusive. Thus, the present study aimed at investigating the protective effects of PZH against CCl₄-induced hepatic fibrosis in rats and exploring the underlying mechanism.

CCl₄-induced hepatic fibrosis is a well-established and the most

commonly used toxin-based experimental animal model for the inducing hepatic fibrosis (Yanguas et al., 2016). When CCl₄ enters the body, it is biotransformed through hepatic cytochrome P450 and yields free radicals that contribute to necrosis of centrilobular hepatocytes, activation of Kupffer cells, and induction of the inflammatory response as well as activation of HSCs, eventually leading to hepatic fibrosis (Constantinou et al., 2005). In this study, CCl₄ causes severe damage to liver tissue as well as repair response of injury in rats, while these pathological lesions were reversed significantly following treatment with PZH, indicating that PZH had an excellent protective effect from hepatic fibrosis.

Inflammation is deemed to be the key promoter that drives this maladaptive wound-healing response in the pathogenesis of CCl₄-induced hepatic fibrosis via the activation of Kupffer cells and release of proinflammatory cytokines, such as TNF- α , IL-1 and IL-6 (Algandaby et al., 2016; Czaja, 2014; Swki and Schwabe, 2016). Intriguingly, NF- κ B is a key transcriptional regulator of the inflammatory response in many hepatic cell populations. A number of inflammatory mediators have been shown to be either targets or activators of NF- κ B. For example, TNF- α and IL-1 β promote NF- κ B activation, and NF- κ B enhances generation of additional TNF- α and IL-1 β . This feedback loop leads to a sustained inflammatory response that ultimately leads to structural impairment and dysfunction in the liver (Rep, 2016; Tsutsui et al., 2015). Notably, in most resting cells, NF- κ B are sequestered in the cytoplasm by inhibitor of κ B (I κ B) proteins, of which I κ B α is the most prominent and masks the nuclear localization and DNA binding domains of the NF- κ B dimer. Upon receptor stimulation, I κ B α is rapidly phosphorylated and triggers ubiquitination and degradation, enabling

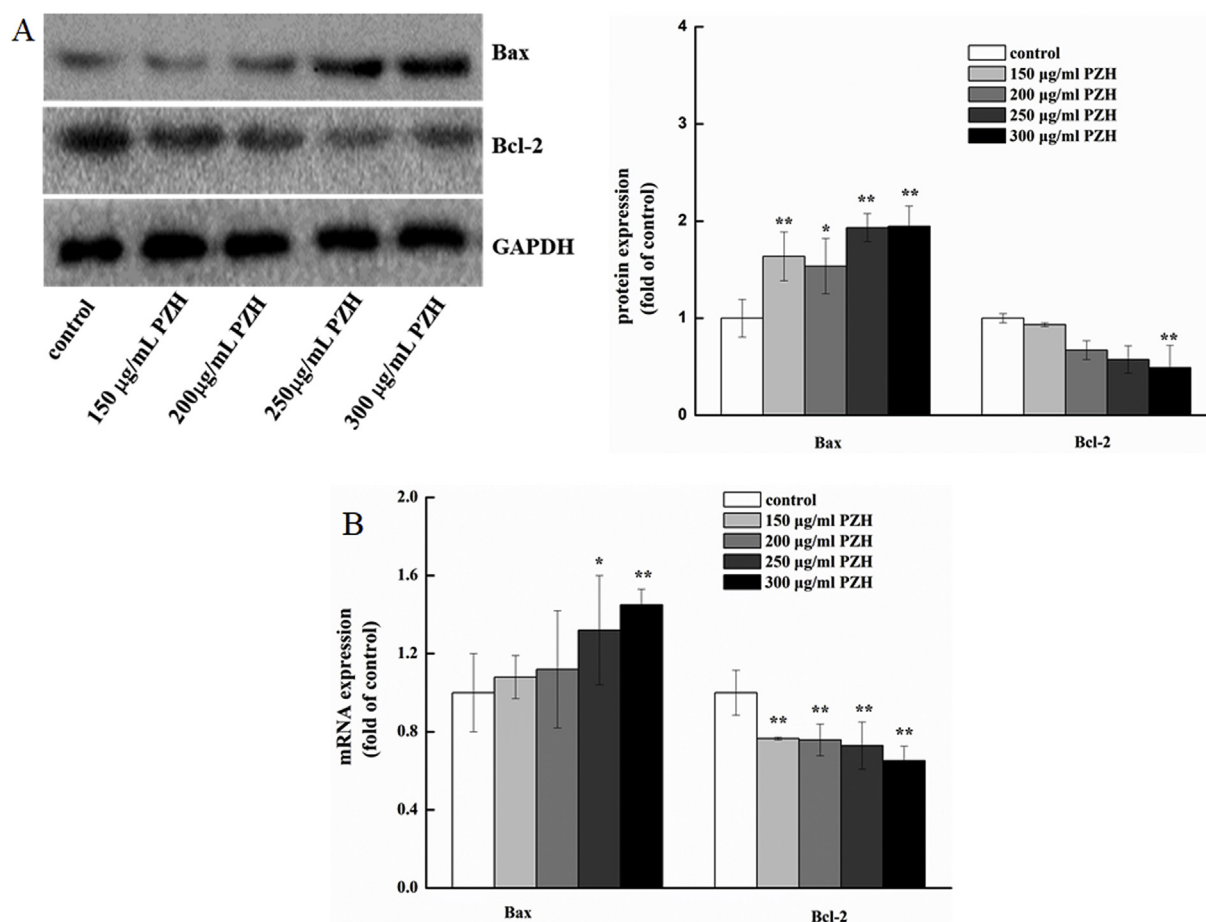


Fig. 8. Effect of PZH on Bcl-2 family in HSC. (A) The expression of Bcl-2 and Bax protein was detected using western bolt assay. (B) The expression of Bcl-2 and Bax mRNA was examined by quantitative PCR assay. Expression was normalized against an endogenous control GAPDH and results were expressed as mean $\pm$ SD of three independent experiments. *p < 0.05, **p < 0.05 vs. the control group.

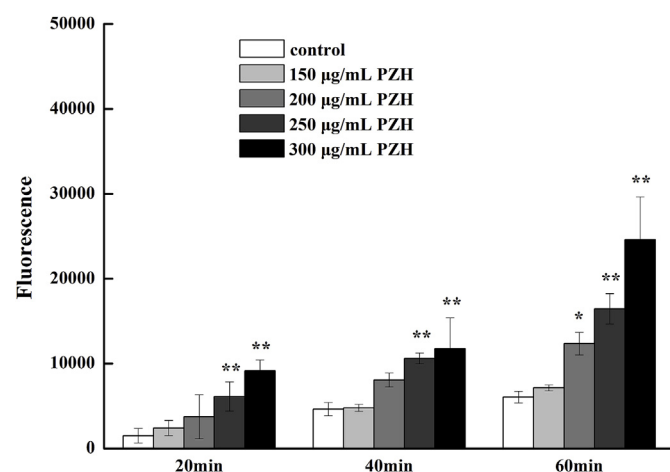


Fig. 9. Effect of PZH on Caspase-3 activity in HSC. *p < 0.05, **p < 0.05 vs. the control group.

nuclear transport of free NF- κ B dimer and subsequently regulation of transcription of target genes. The pivotal signal transduction event in the canonical pathway is therefore phosphorylation of I κ B α , and this is catalyzed by the IKK complex. This complex comprises two catalytic subunits, IKK α and IKK β , and a regulatory subunit. IKK β appears to play the predominant role in the canonical pathway and as such is a strong target for the development of anti-inflammatory drugs (Wang et al., 2014; Zhang et al., 2016). In the present study, we found that the

amount of TNF- α , IL-1 β and IL-6 were dramatically increase in the CCl₄ model group but were markedly mitigated in the PZH-treatment group and colchicine-treatment group. Moreover, CCl₄ significantly upregulated the expression of NF- κ B and its kinase, I κ B- β , as well as down-regulated the expression of the inhibitor, I κ B- α . However, PZH treatment enabled to reverse these situations, suggesting that PZH could alleviate inflammation partially via suppressing the NF- κ B signaling pathway.

For the past decades, it has been demonstrated that the survival and activation of HSCs is a central event in the progress of hepatic fibrosis and HSCs apoptosis is associated with reversible hepatic fibrosis (Chen et al., 2014; Cui et al., 2010; Kim et al., 2013). The mitochondrion-dependent pathway is the most common apoptotic pathway in mammalian cells. This pathway mainly modulated by the Bcl-2 family proteins, which are divided into two groups: pro-apoptotic (Bax, Bad, etc.) and anti-apoptotic (Bcl-2, Bcl-xL, etc.) proteins (Wang, 2015). Therefore, the ratio of pro-apoptotic to anti-apoptotic proteins is potential to determine the fate of cells. Besides, Caspase-3, a key executioner of apoptosis also involved in this process. Of note, several studies have shown that HSC apoptosis is accompanied with a reduction in the Bcl-2/Bax ratio as well as an increase in caspase-3 activity (Chen et al., 2014; Wei et al., 2018). In this study, there was a significant increase in the ratio of HSC-T6 cell apoptosis in a concentration-dependent manner after the treatment with PZH through flow cytometry analysis and morphological observation of DAPI-stained nuclei. Furthermore, we found that PZH could reduce the Bcl-2/Bax ratio, in turn, elevate the caspase-3 activity. These findings indicated that PZH could induce HSC apoptosis primarily via regulation the expression of Bcl-2 family proteins

and caspase-3 activity. On the other hand, it has been proven that multiple transcriptional factors modulate the survival or activation of HSC. Meanwhile, NF- κ B acts as a multifunctional regulator, involved in regulating inflammation, cell survival and liver homeostasis. NF- κ B activation is associated with the transdifferentiation of HSCs into scar-forming hepatic myofibroblasts, an event that is considered pivotal in the fibrogenic response. NF- κ B is activated in HSCs in response to CCl₄-induced injury and stimulates expression of proinflammatory molecules and antiapoptotic factors required for HSC function and survival during the fibrogenic response (Chen et al., 2014; Wang et al., 2017). The present study showed that with the increase of PZH treatment concentration, the expression of NF- κ B in HSCs gradually decreased whilst its inhibitor, I κ B α , accordingly increased. Collectively, these results suggested that PZH could suppress activation or promote apoptosis of HSCs through inhibiting NF- κ B signaling pathway.

In conclusion, this study demonstrates that PZH may attenuate liver fibrosis in rats by inhibiting the NF- κ B pathway, reducing inflammation, and promoting HSC apoptosis. Thereby, PZH is more likely to be a promising antifibrotic agent in chronic disease.

Conflicts of interest

All authors declare that there are no conflicts of interest and have read and approved the submitted manuscript.

Author contributions

Wei X and Haiyin Z conceived and designed the experiments; Xiaoying W, Yuqin Z, Li C and Liping H performed the experiments; Haiyin Z, Yuqin Z and Liping H analyzed the data; Haiyin Z and Yuqin Z wrote the manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2019.111856>.

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